Friday, March 26, 2021
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COS 484: Natural Language Processing

Project Proposal

Paper Name: Scaling Hidden Markov Language Models - Justin T. Chiu & Alexander M. Rush

Link to paper: https://arxiv.org/pdf/2011.04640.pdf

Baselines (Altering Hyperparameters):

- This paper discusses the importance of using Kaiming uniform initialization to initialize weights
- We want to change the weights in three separate ways in order to either improve on their results or represent the importance of initializing weights
- Our three initializations are as follows:
    a. Zero-Initialization - This method will set all of the weights to zero, highlighting the downfall of constant initialization
    b. Random-Initialization - This method will randomly pick the values of the weights using the NumPy random module
    c. Xavier Initialization - This method will use the popular initialization method proposed by by Xavier Glorot and Yoshua Bengio

Codebase: https://github.com/harvardnlp/hmm-lm

Estimate of Timing:

- FF Model: 3 sec/epoch * 100 = 300 sec
- LSTM Model: 23 sec/epoch * 50 = 1150 sec
- VLHMM Model: 433 sec/epoch * 50 = 22150 sec
- Total: 6 hours, 33 min using RTX 2080 GPU

The paper used a RTX 2080 GPU to get the timing above. As a result, we'll need some outside technology to get under 7 hours; but even if we can't, this is still significantly faster than any of the other papers we considered. Our change in how the weights are initialized might affect the timing of each experiment. We will run each model simultaneously on different systems and be cognizant of the fact it might take longer to complete each experiment.

Proposed Timeline and Contributions:

Week of March 21st - Complete Project Proposal (**Thomas, AJ, Caroline**)
Week of March 28th - Prepare technology being used (**AJ**) reproduce paper (**Thomas, Caroline**)
Week of April 4th - Zero (**Caroline**) Random (**AJ**) Xavier (**Thomas**) implementations
Week of April 11th - Compile and interpret results of implementations (**Thomas, AJ, Caroline**)
Week of April 18th - Write paper:
Abstract, Introduction, Section on Related Work, Xavier Analysis (**Thomas**)
Proposed Idea, Technical Details, Random Analysis (**AJ**)
Baseline Comparison, Citations, Zero Analysis (**Caroline**)
Week of April 25th - Compile and format paper, practice presentation (**Thomas, AJ, Caroline**)
Week of May 2nd - Finalize presentation and turn in paper (May 5th) (**Thomas, AJ, Caroline**)